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EXPERIMENTAL MEASUREMENT OF BOUNDARY LAYER FLOW WITH SUCTION INTO A HONEYCOMB PANEL

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ABSTRACT

Experiments were conducted in a low speed laboratory wind tunnel in order to characterize boundary layer flow that is subjected to suction as it flows over a honeycomb section of the wall. The problem arises in the design of filtered inlets to engines in aerodynamic applications. The honeycomb was baselined in a traditional orientation normal to the flow direction. The overall total and static pressure drops and the downstream velocity field were measured using pressure probes and wall static taps upstream and downstream of the honeycomb. The wind tunnel was then re-configured so that the main wind tunnel flow resembled an infinite free-stream approaching the honeycomb inlet at 90° from the normal. Total and static pressure surveys were performed in the airflow upstream of the honeycomb, and at the honeycomb inlet and exit. Single hot wire anemometer surveys of the time mean total velocities were performed at the honeycomb inlet and at various locations downstream of the honeycomb inlet. Because the honeycomb exit flow has a single velocity component, the measured inlet and exit total velocities were sufficient to determine the local approach angle of the inlet velocity vector. It was determined that the radius of curvature of the flow stream entering the honeycomb is small in the upstream region of the honeycomb, and increases monotonically in the flow direction. This largely inviscid effect causes a drop in the static pressure in the upstream region of the honeycomb followed by a gradual recovery of the static pressure in the flow direction. The local mass flow ingested through the honeycomb is low in

the forward section and increases monotonically in the flow direction. This non-uniformity in the ingested mass flow results from the non-uniformity in the turning radius and entering static pressure, an inviscid mechanism, and from non-uniformity of the pressure loss coefficient over the honeycomb panel, a viscous mechanism. The total pressure loss coefficient, K , is found to vary from a value of roughly 1.2, for grazing angle of 5° in the forward section of the honeycomb, to 0.9, for a flow approaching from the direction normal to the inlet. It is hypothesized that the increased total pressure loss for the grazing flow is due to flow separation at the inlet of each honeycomb passage.

NOMENCLATURE

D	honeycomb cell diameter
f	friction factor
H	height of test section
K	loss coefficient $\Delta P / (\frac{1}{2} \rho V^2)$
K_S	loss coefficient based on static pressure $\Delta P_S / (\frac{1}{2} \rho V^2)$
K_T	loss coefficient based on total pressure $\Delta P_T / (\frac{1}{2} \rho V^2)$
L	length of honeycomb panel
$P_{s,0}$	static pressure at station 0
$P_{s,1}$	static pressure at station 1
$P_{s,2}$	static pressure at station 2
$P_{t,0}$	total pressure at station 0
$P_{t,1}$	total pressure at station 1
$P_{t,2}$	total pressure at station 2

R	flow turning radius
Re	Reynolds number
Re_D	Reynolds number based on cell diameter
Re_L	Reynolds number based on length of honeycomb
SR	suction ratio V_2/V_0
t	honeycomb panel thickness
V_0	velocity at station 0
V_1	velocity at station 1
V_2	velocity at station 2
V_D	suction duct velocity
W	honeycomb panel width
X	distance along honeycomb
Y	distance in front of honeycomb
Y'	distance behind honeycomb
α	porosity of honeycomb
θ	grazing angle
ρ	flow density
ρ_D	suction duct density
ρ_∞	free-stream density

INTRODUCTION

With an increased emphasis on installed power capabilities, the design of helicopter inlets has become more challenging. Simple forward-facing inlet openings are no longer acceptable. Many different inlet styles have been proposed that meet the new engine inlet requirements. One of these inlet styles incorporates the use of a honeycomb material that is arranged into panels that are installed flush with the helicopter fuselage. The inlet duct that feeds the gas turbine engine attaches to the downstream side of these honeycomb panels. The honeycomb can effectively protect the engine and enhance performance under some conditions

While effective as engine inlet protection, these flush honeycomb panels prove to be an aerodynamic design challenge. This is especially true during forward flight conditions. In these conditions, the air that feeds the helicopter's gas turbine engines flows along, or grazes, the fuselage and flush honeycomb panels and must make a dramatic 90° turn into the honeycomb before entering the engine through the inlet duct. This dramatic change in airflow direction can result in a significant increase in total pressure drop through the honeycomb, compared to the situation in which the flow approaches from a direction normal to the inlet plane. Additionally, if the honeycomb introduces non-uniformity in the airflow pattern, this can affect the total pressure and swirl distortion at the compressor eye of the gas turbine engine. This increased pressure drop and distortion can significantly impact the performance and operability of the gas turbine engine.

In order to capture the aerodynamic impact of the honeycomb style inlet system, Computational Fluid Dynamics (CFD) models are commonly used to model the external approach flow, the honeycomb, and the downstream inlet duct feeding the engine. The use of honeycomb panels introduces challenges and uncertainties into the CFD modeling. Creating a local grid which models the honeycomb structure would result in grid sizes that are prohibitively large. An alternative approach is to model the honeycomb as a Porous Media volume element where the pressure drop that represents the impact of the honeycomb panel is calculated using simple algebraic formulations resulting from the theory of porous media. In that formulation, the pressure loss is ultimately computed by locally applying total pressure drop coefficients, K . Empirical loss coefficients for flow through honeycomb like structures exist in the literature, but only for flow directed normally at the honeycomb [1]. No data exists in the open literature for the pressure drop through honeycomb exposed to a grazing flow.

A wind tunnel test was devised to accurately measure the pressure drop and airflow pattern of a honeycomb panel exposed to a grazing flow. The situation is shown in Figure 1. The goal of the measurements was to provide a benchmark experiment for validation of the CFD approach, and in particular to determine if honeycomb pressure loss coefficients for flow approaching in a direction normal to the honeycomb inlet are acceptable for inlet design in a grazing flow situation. Because honeycomb is by nature a flow straightener, it was determined during the course of the measurements that not only was the experiment suitable as a validation experiment, but it could also provide useful information on the dependence of pressure loss coefficients on the angle of the grazing flow vector. This fortunate circumstance allowed a quantitative comparison of local loss coefficients in normal flow to those in grazing flow.

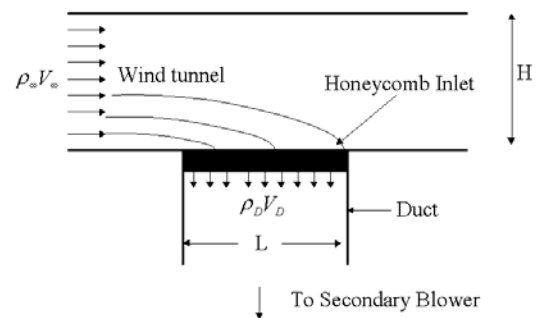


Figure 1. Conceptual View of Wind Tunnel Experiment Simulating Grazing Flow at a Honeycomb Inlet

APPARATUS AND INSTRUMENTATION

Experiment Apparatus

The test article was a hexagonal honeycomb panel that had a streamwise length to spanwise width ratio, L/W of 3. The panel had a thickness to a cell diameter ratio, t/D of approximately 9 and a porosity of 86%. This honeycomb panel was tested in two configurations, flow approaching normal and parallel (grazing) to the honeycomb panel in a low speed, open return wind tunnel. The wind tunnel is an existing research quality open return wind tunnel that has been qualified and used many times for other tests. The main test section is $4L$ long, $0.3L$ across the span, and L in the direction normal to the honeycomb (Figure 2.). The flow enters the test section after passing through a flow management section and an 8:1 contraction that produces a flow with $\pm 0.1\%$ velocity uniformity and 1% RMS turbulence intensity.

For flow approaching normal to the honeycomb, the panel was secured 104 cell diameters downstream of a wind tunnel contraction. The honeycomb panel filled the entire test section. A 0.125 -inch diameter Dwyer Model 166-12 pitot tube was placed such that the tip was 48 cell diameters upstream of the honeycomb panel to measure the inlet flow velocity. Downstream of the honeycomb panel, a probe holder was affixed to a motorized traverse system so that probes could be traversed at various vertical and downstream locations behind the honeycomb. Along the centerline of the side of the test section a series of static wall taps were inserted in increments of 32 cell diameters for a total length of $224D$.

In order to calculate the loss coefficient based on the total pressure drop for the normal flow orientation, a total pressure probe with a tip diameter of $0.28D$ located at $Y'/D = 16$ was used. The total pressure probe was traversed in increments of half a probe diameter in the X -direction for a length of $0.33L$. As will be seen in a later section, local non-uniformities in the total pressure occurred as a result of the individual jets issuing from each honeycomb channel. From this total pressure survey an average total pressure downstream of the honeycomb was calculated. The total pressure upstream of the honeycomb was determined by using the total pressure port on the upstream pitot tube.

For the grazing flow orientation, the wind tunnel was re-configured with a secondary discharge duct with suction provided far downstream by a secondary centrifugal blower. The honeycomb panel was inserted into the inlet of the secondary duct so that its upper surface was flush with the wind tunnel surface (Figure 3). The suction side test duct was located 0.26 panel length downstream of the main wind tunnel contraction exit. The suction duct was one panel length wide, 0.92 panel length long and 0.33 panel length deep. Figure 3 shows the locations of the static wall taps fitted on one of the sidewalls in the main flow duct. In Figure 3, key regions of interest are identified that will be referred to in the results and

discussion section of this paper. These regions are as follows: station 0 is the approach condition located 0.25 panel length upstream of the honeycomb panel, station 1 is the region one cell diameter in front of the honeycomb panel, and station 2 is the region one cell diameter behind the honeycomb panel.

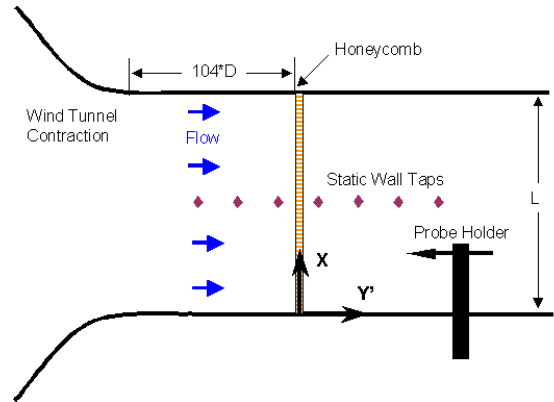


Figure 2. Wind Tunnel Configured for Measurements of Flow Approaching From Normal Direction

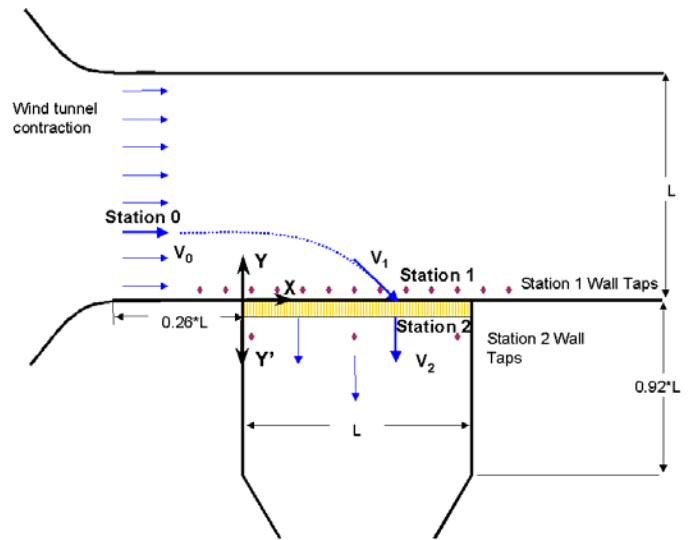


Figure 3. Details of Grazing Flow Wind Tunnel Configuration

Instrumentation

Measurement of pressure from static wall taps, pitot tubes, and total pressure probes were performed with a common set of instruments. All pressure-sensing devices were fitted with 0.125 -inch diameter Tygon tubing that were attached to a 48-channel scanivalve. A MKS Baratron 698A pressure transducer read the selected channel on the scanivalve. The MKS Baratron 698A pressure transducer was equipped with a MKS Type 270 signal-conditioning unit. The output voltage

was read through a PCI 6071E National Instruments data acquisition card fitted inside of a desktop PC.

A Model IFA 1210 TSI single wire hotwire anemometer probe connected to a TSI model IFA 100 hotwire anemometer controller was used to measure the velocity at various locations in the test section. The hotwire probe was calibrated in the exit of a thermally controlled vertical air jet against the static pressure at the settling chamber with a MSK Baratron 698A pressure transducer. The hotwire probe was calibrated at different velocities but only at one air temperature. The output voltage of the TSI model IFA 100 was measured with the National Instruments PCI 6071E data acquisition card. The voltage and velocity calibration data was curve fitted with a fourth-order polynomial. In order to account for small air temperature fluctuations, an algebraic correction factor was used as suggested by the manufacturer [2].

When calculating the air velocity from total pressure, the air density was calculated by measuring the air temperature by using a Fluke Hydra 2620A data acquisition unit and the room pressure by a wall mounted Sargent-Welch Scientific Company S-4519 mercury barometer.

Experiment Uncertainty

The exit condition of the main wind tunnel had uniform velocity profile of $\pm 0.1\%$ of the mean velocity. It also had a temporal drift of less than $\pm 0.1\%$ of the mean velocity. The uncertainty of the MKS Baratron 698A pressure transducer was $\pm 0.2\%$ of the reading. The hotwire velocity calibration uncertainty was $\pm 1.2\%$ of the reading. This uncertainty was found by comparing the fourth-order polynomial calibration curve against the static pressure of the vertical air jet's settling chamber in the same flow as the hotwire was calibrated.

RESULTS

Velocity and Pressure Behavior for Normal Approach Flow

One of the first objectives in the normal flow configuration was to determine the velocity profile behind the honeycomb panel. Figures 4.a, 4.b and 5 show the total time-averaged velocity measured by using a total pressure probe with a tip diameter of $0.28D$ and static wall taps for the normal flow cell Reynolds number of 730 and 2300. The probe was traversed in the X-direction in increments of half the probe diameter over $0.5L$ long surveys at three Y' locations: $Y'/D = 4$, 16, and 96. Figure 4.b shows that the large periodic non-uniformities occur at wavelengths consistent with the honeycomb cell diameter, most certainly due to individual jets issuing through the honeycomb cells. The jets are quite coherent, and persist to at least $Y'/D = 16$ downstream. At $Y'/D = 96$, the individual jets have merged and non-uniformities do not exceed 2% of the

time-averaged mean velocity. The observed remnant non-uniformities in the velocity profile are probably due to imperfections in the honeycomb, rather than to any jet induced effects. The local total pressure measurements certainly point out that total pressure measurements at a single location behind a screen or honeycomb are unsuitable for characterizing the time and space averaged behavior.

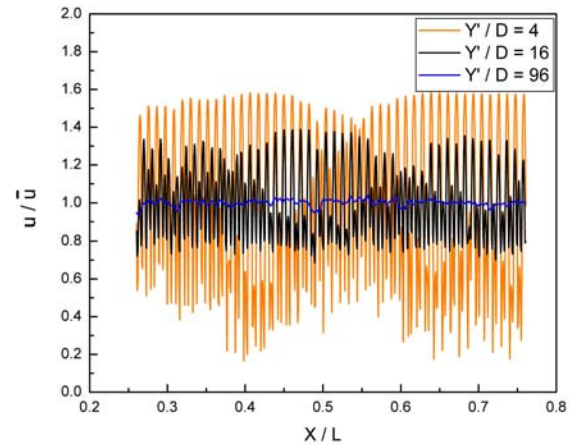


Figure 4. a. Time Averaged Velocity Downstream of the Honeycomb for Normal Flow at a cell Reynolds Number of 730

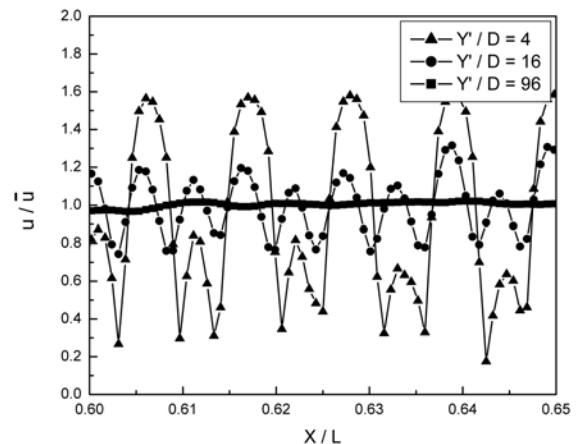


Figure 4. b. Detail View of figure 4.a, Time Averaged Velocity Downstream of the Honeycomb for Normal Flow at a cell Reynolds Number of 730

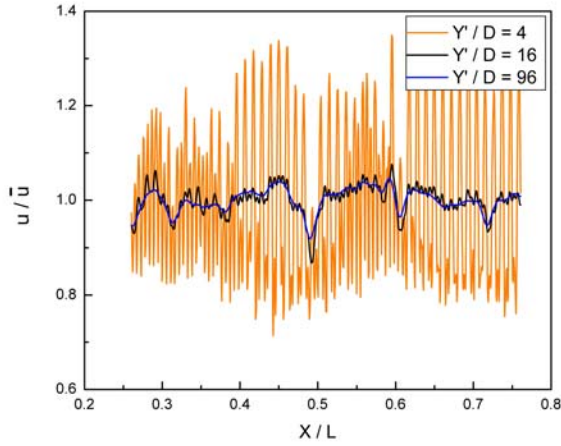


Figure 5. Time Averaged Velocity Downstream of the Honeycomb for Normal Flow at a cell Reynolds Number of 2300

The loss coefficients in Figure 6 are both the static pressure loss coefficient, K_S and the total pressure loss coefficient, K_T . K_S was arbitrarily defined using the difference between the wall tap located at $Y'/D = -80$ and the wall tap located at $Y'/D = 40$. The total pressure loss coefficient, K_T , was defined using the total pressure port of the pitot tube at $Y'/D = -48$ and the average total pressure from a survey at $Y'/D = 16$. The approach velocity was measured 48 cell diameters upstream. As seen in Figure 6, even though the loss coefficient can be defined either on static or total pressure drop, they are approximately the same value because the dynamic pressure component is the same on both sides of the honeycomb, due to mass continuity and constant cross-sectional areas. A possible explanation for the small difference between the two is that the effects of upstream stagnation behavior in the static pressure and static pressure recovery at the exit were not completely captured when measuring the static pressure drop across the honeycomb for all the different Reynolds numbers investigated. Because of these effects, it clear why the loss coefficient is commonly defined on the total pressure since these effects will not occur in the total pressure. Also plotted in Figure 6 is the analytical solution for the loss coefficient of an assemblage of small passages. At low Reynolds number the behavior has strong Reynolds number dependence; the dependence decays as Reynolds number increases beyond 1000, as the internal flow in the passages becomes turbulent.

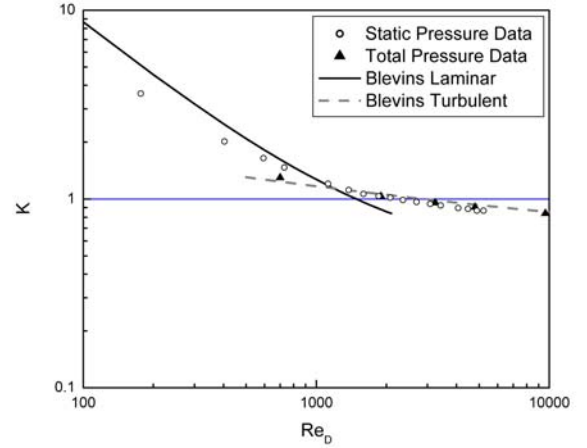


Figure 6. Loss Coefficient for Normal Flow

ANALYTICAL COMPARISON

The analytical solution used in Figure 6 was the solution for a loss coefficient of a grillage or perforated plate with square edges by Blevins [1]. The loss coefficient is given by:

$$K = \frac{\frac{1}{2}(1-\alpha) + (1-\alpha^2) + \frac{f^*t}{D}}{\alpha^2} \quad (1)$$

In Equation 1, K is the loss coefficient, α is the porosity of the material, f is the friction factor, t is the thickness of the material parallel to the flow direction and D is the cell diameter.

Blevins [1] leaves off the viscous term $\left[\frac{f^*t}{D}\right]$ for $D < t <$

50D. However, it was found that this term was necessary to obtain agreement with the present data and should not be neglected for this honeycomb structure. The accuracy of this solution is stated to be $\pm 20\%$ with large errors possibly occurring $\alpha > 0.85$. The friction factor in the laminar regime ($Re < 1000$) is calculated using the standard formulation for fully-developed laminar flow in tubes:

$$f = \frac{64}{Re} \quad (3)$$

In the turbulent regime the friction factor is calculated with the Blasius correlation for turbulent flow in smooth pipes [1], valid for $4 \times 10^3 < Re < 2 \times 10^5$.

$$f = \frac{0.316}{Re^{0.25}} \quad (2)$$

The analytical results are on the average 5% different from the experimental data, except at low Reynolds numbers ($Re < 700$) where there is much higher error. This would suggest that for normal flow, the current analytical expression for the loss

coefficient is sufficient; in particular it suggests that the inlet and exit loss coefficients are satisfactorily predicted.

VELOCITY AND PRESSURE BEHAVIOR FOR GRAZING FLOW

Flow Visualization

Before any detailed measurements were taken for the grazing flow orientation, a string tuft matrix was constructed in front of and behind the honeycomb panel. The flow visualization revealed the region in which the bulk of the flow turned into the honeycomb. It also showed that the approaching main flow, although confined by a duct, had only minor streamwise velocity gradients and hence appeared to resemble a constant velocity free-stream flow to the honeycomb inlet.

A TSI two-camera particle image velocimetry (PIV) system was briefly used to test a small region in front of the honeycomb panel. The results showed that the flow had very small spanwise velocity components and thus could be assumed to be a two-dimensional flow along the streamwise centerline.

Detailed Measurements

Detailed grazing flow measurements were made at a single test condition. The approach flow had a Reynolds number based on the panel length of 10^6 , and a cell Reynolds number of 5500. At station 2 the exit flow had an average Reynolds number based on the panel length of 10^5 , and a cell diameter Reynolds number of 570. The suction ratio, SR, was 10.8%.

Figure 7 is a plot of the total velocity profile at station 1, one cell diameter in front of the honeycomb panel measured with a single wire hotwire anemometer. The error bars plotted on the data set are not the uncertainty in the data, but rather the local measured rms turbulence. The data show that even though the flow undergoes turning, the total velocity remains approximately the same as the approach velocity at station 0. Most remarkable is the region just downstream of the honeycomb panel where there is a steep dip in the velocity followed by a rapid recovery. A possible explanation for this behavior is the presence of a dividing streamline and a stagnation point just after the honeycomb panel. This would cause the flow that does not get sucked into the honeycomb to behave as though it was entering an expanding duct. If this expansion is abrupt, it is possible that a recirculation region will form in the flow at the expansion. This may explain why the total velocity dramatically decreases, followed by an abrupt recovery.

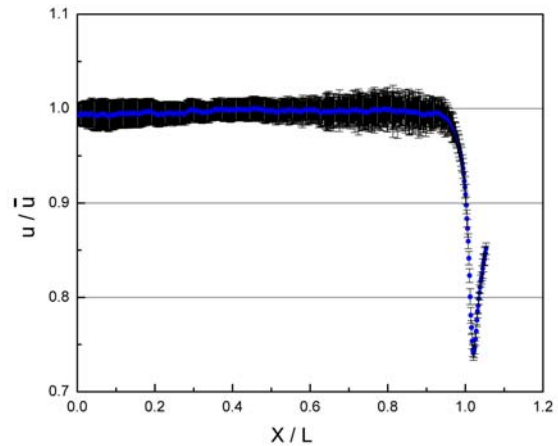


Figure 7. Total Velocity Profile one cell diameter in Front of Honeycomb for Grazing Flow

Figure 8 is a plot of the total velocity profile at $Y'/D = 1$, and 57 behind the honeycomb panel measured with a single wire hotwire anemometer. Because the honeycomb is a natural flow straightener, the lateral velocity component vanishes, and thus the velocity profile of Figure 8 also represents the normal velocity component that passes through the honeycomb. The most obvious trend is that the mass ingested through the honeycomb is low in the forward (upstream) section and monotonically increases in the streamwise direction toward the aft section. The data show some of the non-uniformities associated with the individual cell jets, especially in the center third of the honeycomb panel. Surprisingly, in the forward and aft thirds, the flow is more uniform than would be expected when compared to the normal flow data. A fifth order polynomial was fit through the data at one cell diameter behind the honeycomb. This approximated velocity profile can be used to compare to CFD results since CFD results will not capture the individual jet effects.

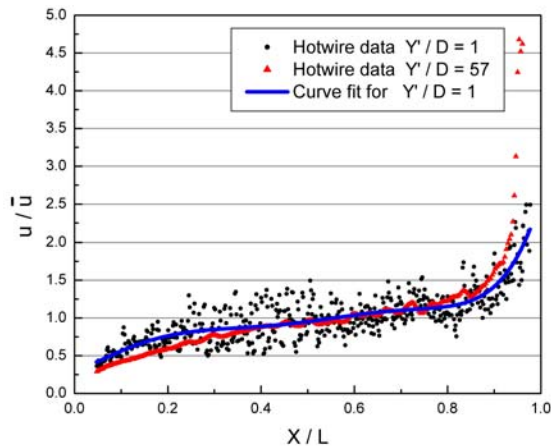


Figure 8. Total Velocity Profile behind Honeycomb for Grazing Flow

Figure 9 is a plot of the grazing angle at station 1. Since the honeycomb panel straightens the flow, the measured total velocity at station 2 in Figure 8 is indeed the normal component of the total velocity at station 1. Since the total and normal components are known, the main flow streamwise component and the grazing angle were calculated. An important observation to note is that the grazing angle entering the honeycomb panel is small, with a maximum value of about 15° at the aft portion of the inlet.

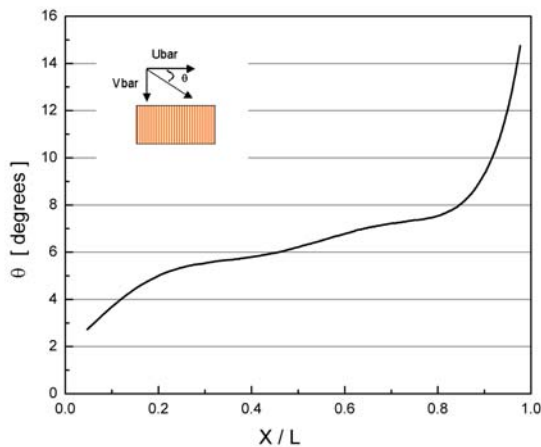


Figure 9. Grazing Angle at One Cell Diameter in Front of Honeycomb for Grazing Flow

The local static pressures at stations 0, 1, and 2 are shown in Figure 10, and the pressure differences are shown in Figure 11. The plotted value at station 0 is the static pressure measured with the static port of a pitot tube. It is assumed that there is no static pressure variation at station 0 since the flow is uniform and unidirectional. The static pressure at station 1 is the static

pressure measured from static wall taps located one cell diameter in front of the honeycomb panel. The static pressure at station 2 was measured with a straight 0.0625-inch pitot tube with the static ports located 4 cell diameters behind the honeycomb panel. The heavy line through the data at station 1 is a polynomial curve fit was used to calculate the pressure differences in Figure 10. Figure 11 shows that the static pressure drop due to inviscid turning is $2/3$ of the overall pressure drop from station 0 to station 2. The static pressure at station 1 has the same shape as the polynomial curve fit of the velocity data in figure 10. Without a doubt the local static pressure drop across the honeycomb panel produces the velocity profile behind the honeycomb since a static pressure difference drives the flow. Since the static pressure boundary condition on the backside of the honeycomb is uniform and constant, then it is clear that the local mass ingested through the honeycomb is set, at least partly, by the degree of turning that the flow undergoes. This appears to be a purely inviscid effect. This data also shows that the goal of establishing a nearly constant static pressure condition behind the honeycomb was achieved.

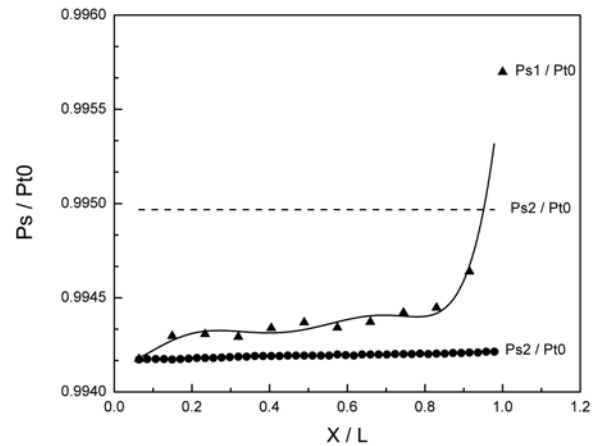


Figure 10. Local Static Pressure at Various Stations for Grazing Flow

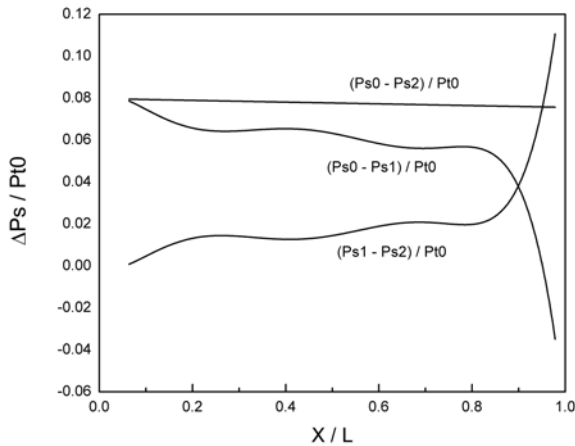


Figure 11. Static Pressure Drop for Grazing Flow

Figure 12 shows the local total pressure at stations 1 and 2. The total pressure at station 1 was measured with a 0.125-inch United Sensor model KBC-12 kiel probe that could accommodate an approach angle of ± 45 degrees. The total pressure at station 2 was measured with a custom total pressure probe with a tip diameter of $0.28D$. The figure shows that even though there is a total pressure variation at station 2, it is not significant when compared to the overall total pressure drop across the honeycomb panel. In fact, the overall pressure drop across the honeycomb panel can be estimated by a single value without a large percent of error. Figure 12 in addition shows that there is virtually no total pressure loss from station 0 to station 1. This suggests that the previous observation made from the data in Figure 11, that the static pressure drop from station 0 to station 1 is purely an inviscid effect due to the flow turning, is correct.

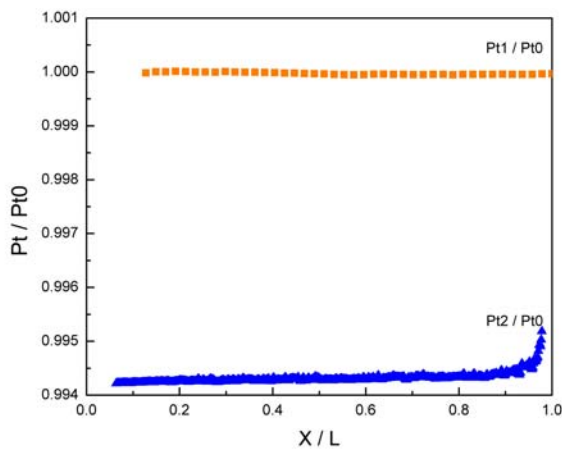


Figure 12. Total Pressure at Station 1 and Station 2

Total Pressure Loss Coefficients

From the outset, one of the goals of the experimental study was to assess the impact of flow turning on the total pressure loss coefficient, K_T . The data of Figure 9 and 12, as well as the data for the normal flow configuration, Figure 6, allowed the comparison of the total loss coefficient measured in grazing flow to that in normal flow, Figure 13. It is seen that the loss coefficient decreases monotonically with approach grazing angle from 1.15 at 5° to 1.05 at 15° , where the grazing angle is measured from the line parallel to the honeycomb inlet plane. In contrast, the loss coefficient asymptotes to a value of 0.9 in the normal flow situation where the grazing angle is 90° . Data in the region between 15° and 90° was not obtained, as this grazing angle was not achieved with the suction ratio (SR) of 10% that these data represent.

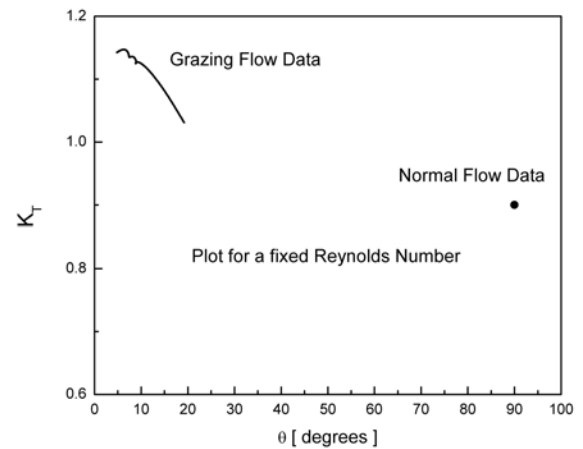


Figure 13. Loss Coefficient for Normal Flow and Grazing Flow

DISCUSSION

The data of Figure 13 allow for the formation of a working hypothesis as to the differences between the grazing and normal flow situations and their impact on the pressure loss coefficient. Figure 14 depicts a possible behavior. In this scenario the flow turns as it approaches the honeycomb and the static pressure is established by the radius of streamline curvature, possibly a purely inviscid turning mechanisms. As the measured data illustrate, the static pressure drops dramatically in the upstream portion of the honeycomb, where the grazing angle (and streamline curvature) is severe and recovers towards the free-stream condition as the grazing angle decreases. The flow does not enter the honeycomb inlet plane normally, rather it enters at an angle that depends on the suction ratio and probably the approach Reynolds number. Figure 14 elaborates this hypothesis in an exaggerated manner. Because of the grazing angle, flow separation occurs at the honeycomb inlet. This perturbation on the inlet conditions is

greatest under grazing flow ($\theta = 0^\circ$) and relaxes as the flow approaches the normal flow configuration. The flow separation is a viscous and irreversible mechanism that does indeed result in an irrecoverable total pressure loss.

In summary, the total pressure drop from 0 to 2 is deceptively complex. From the freestream (0) to the inlet plane (1), there is virtually no total pressure drop but the static pressure drops in proportion to the severity of the turn into the inlet plane of the honeycomb, that is, in inverse proportion to the radius of curvature of the approaching streamline. The experiments confirm that the static pressure drops most severely at the most forward section of the honeycomb, then gradually recovers in the flow direction, consistent with this inviscid hypothesis. Second, the loss coefficient increases in proportion to the degree of flow separation that occurs at the inlet, and as illustrated in Figure 14, the separation is in proportion to the grazing angle at the inlet plane. As a consequence, the flow is starved in the forward section of the honeycomb inlet for two reasons: first, the static pressure at the inlet plane is reduced because of severe turning, and second, the severe turning results in flow separation and an increase in the flow resistance through the honeycomb passage, compared to a normal approach flow.

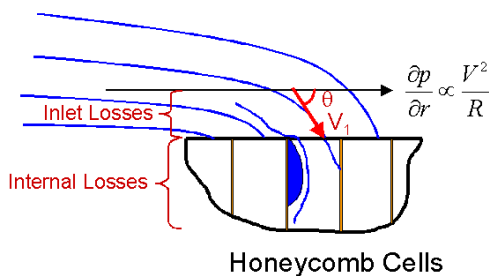


Figure 14. Sketch of Incoming Flow Behavior

CONCLUSIONS

1. The experimental study produced a set of data for a single mainstream approach Reynolds number and a single suction ratio. The data are consistent and repeatable and can be used for benchmarking a CFD investigation.
2. Under normal approach flow, the honeycomb total pressure loss coefficient was well predicted by literature correlations in the turbulent regime, but was lower than the predicted behavior in the laminar regime.
3. From the free-stream to the inlet plane of the honeycomb, the total pressure remains virtually constant, but the static pressure changes as a result of the streamline curvature that is experienced as the

flow turns to enter the honeycomb. The static pressure significantly decreases in the forward section of the honeycomb, where the turning is severe, and recovers in the flow direction as the streamline curvature decreases.

4. Turning causes the flow to approach the honeycomb with a flow angle. It is hypothesized that the more severe the flow angle, the greater the degree of flow separation at the inlet of each honeycomb passage. Hence, the data show a strong correlation between the loss coefficient, K_T , and the approach flow angle.
5. The combination of inviscid static pressure decrease at the honeycomb inlet, and increased flow resistance as a result of flow separation, has the net effect of starving the honeycomb inlet in the forward section, with the locally ingested flow increasing monotonically in the flow direction.

CONTINUING WORK

The wind tunnel has been modified to produce higher main flow Reynolds number and higher suction velocities. This will provide the needed data to assess Reynolds number and suction ratio dependence on the loss coefficient. Results will also be compared to CFD simulation of the same experimental setup. Simple analytical models are also being developed, possibly based on stream tube analysis with the empirical loss coefficients embedded into the method.

ACKNOWLEDGEMENTS

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